



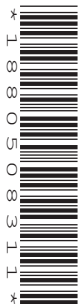
Oxford Cambridge and RSA

Friday 23 May 2025 – Afternoon

AS Level Mathematics A

H230/02 Pure Mathematics and Mechanics

Time allowed: 1 hour 30 minutes



You must have:

- the Printed Answer Booklet
- a scientific or graphical calculator

QP

INSTRUCTIONS

- Use black ink. You can use an HB pencil, but only for graphs and diagrams.
- Write your answer to each question in the space provided in the **Printed Answer Booklet**. If you need extra space use the lined page at the end of the Printed Answer Booklet. The question numbers must be clearly shown.
- Fill in the boxes on the front of the Printed Answer Booklet.
- Answer **all** the questions.
- Where appropriate, your answer should be supported with working. Marks might be given for using a correct method, even if your answer is wrong.
- Give non-exact numerical answers correct to **3** significant figures unless a different degree of accuracy is specified in the question.
- The acceleration due to gravity is denoted by $g \text{ ms}^{-2}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.
- Do **not** send this Question Paper for marking. Keep it in the centre or recycle it.

INFORMATION

- The total mark for this paper is **75**.
- The marks for each question are shown in brackets [].
- This document has **8** pages.

ADVICE

- Read each question carefully before you start your answer.

Formulae
AS Level Mathematics A (H230)

Binomial series

$$(a+b)^n = a^n + {}^nC_1 a^{n-1}b + {}^nC_2 a^{n-2}b^2 + \dots + {}^nC_r a^{n-r}b^r + \dots + b^n \quad (n \in \mathbb{N}),$$

$$\text{where } {}^nC_r = {}_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Differentiation from first principles

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Standard deviation

$$\sqrt{\frac{\sum(x-\bar{x})^2}{n}} = \sqrt{\frac{\sum x^2}{n} - \bar{x}^2} \quad \text{or} \quad \sqrt{\frac{\sum f(x-\bar{x})^2}{\sum f}} = \sqrt{\frac{\sum fx^2}{\sum f} - \bar{x}^2}$$

The binomial distribution

If $X \sim B(n, p)$ then $P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$, mean of X is np , variance of X is $np(1-p)$

Kinematics

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$s = \frac{1}{2}(u+v)t$$

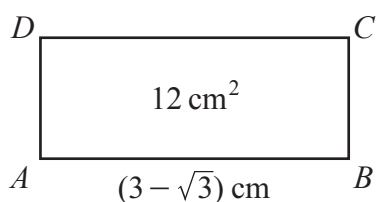
$$v^2 = u^2 + 2as$$

$$s = vt - \frac{1}{2}at^2$$

Section A
Pure Mathematics

- 1 Determine the equation of the line that passes through the point $(1, 3)$ and is perpendicular to the line with equation $3x + 6y - 5 = 0$. Give your answer in the form $ax + by + c = 0$ where a , b and c are integers to be determined. [3]

- 2 In this question you must show detailed reasoning.



The diagram shows the rectangle $ABCD$ where $AB = (3 - \sqrt{3})$ cm.

The area of $ABCD$ is 12 cm^2 .

Find the perimeter of $ABCD$. Give your answer in the form $(a + b\sqrt{3})$ cm, where a and b are integers to be determined. [4]

- 3 In a triangle ABC , $AB = 9$ cm, $BC = 7$ cm and $AC = 4$ cm.

(a) Show that $\cos CAB = \frac{2}{3}$. [2]

(b) Hence find the exact value of $\sin CAB$. [2]

(c) Find the exact area of triangle ABC . [2]

- 4 The cubic polynomial $2x^3 - kx^2 + 4x + k$, where k is a constant, is denoted by $f(x)$. It is given that $f'(2) = 16$.

(a) Show that $k = 3$.

[4]

For the remainder of the question, you should use this value of k .

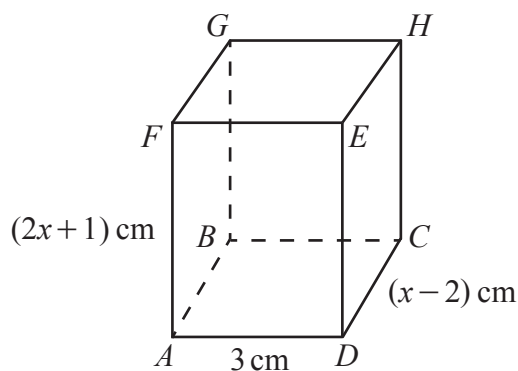
(b) Use the factor theorem to show that $(2x + 1)$ is a factor of $f(x)$.

[2]

(c) Hence show that the equation $f(x) = 0$ has only **one** real root.

[4]

- 5 In this question you must show detailed reasoning.



The diagram shows the cuboid $ABCDEFGH$ where $AD = 3$ cm, $AF = (2x + 1)$ cm and $DC = (x - 2)$ cm.

The volume of the cuboid is at most 9 cm^3 .

Find the range of possible values of x . Give your answer in interval notation.

[5]

- 6 (a) The curve $y = \ln x$ is stretched by a scale factor 3 in the x -direction.

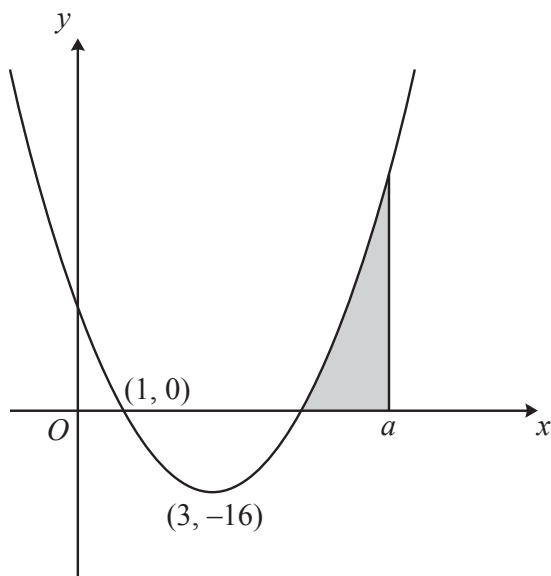
State the equation of the curve after it has been stretched.

[2]

- (b) Describe fully the **single** transformation that transforms the curve $y = x^2 + 1$ to the curve $y = x^2 - 6x + 8$.

[4]

7



The diagram shows the curve with equation $y = f(x)$ where $f(x)$ is a quadratic function. The curve intersects the x -axis at the point $(1, 0)$ and has its minimum at the point $(3, -16)$.

- (a) Show that $f(x) = 4x^2 - 24x + 20$.

[3]

The shaded region is enclosed by the curve, the x -axis, and the line $x = a$. The area of the shaded region is $\frac{100}{3}$ units².

- (b) Determine the exact value of the constant a .

[6]

- 8 The curves $e^x - 2e^y = 1$ and $2e^x + 3e^{2y} = 41$ intersect at the point P .

- (a) Show that the y -coordinate of P satisfies the equation $3e^{2y} + 4e^y - 39 = 0$.

[1]

- (b) In this question you must show detailed reasoning.

Hence find the exact coordinates of P .

[5]

Section B
Mechanics

- 9 An athlete runs with velocity $v \text{ m s}^{-1}$ along a horizontal straight running track AB . At time t seconds after passing A , $v = 0.05t^2 - 0.4t + k$, where k is a constant.

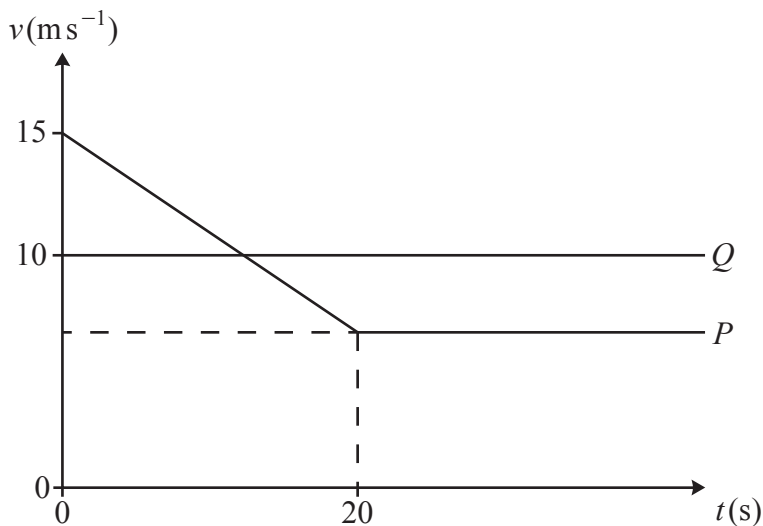
(a) Find the acceleration of the athlete when $t = 6$. [2]

(b) Given that the minimum velocity of the athlete is 6.6 m s^{-1} , show that $k = 7.4$. [3]

The athlete runs in a straight line from A to B in 13.1 seconds.

(c) Calculate the distance AB . [2]

10

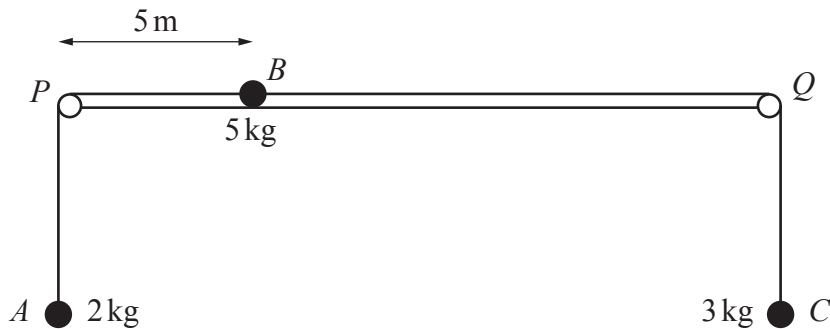


The velocity-time graph represents the motion of two cyclists P and Q who are travelling along a straight, horizontal road. At time $t = 0$, the cyclists pass the same point on the road with P cycling at 15 m s^{-1} and Q cycling at 10 m s^{-1} .

From time $t = 0$, P cycles with constant deceleration 0.4 m s^{-2} for 20 seconds and Q cycles at a constant velocity of 10 m s^{-1} . From $t = 20$ onwards, P 's velocity remains constant.

(a) Find the value of t when P and Q have the same velocity. [2]

(b) Determine the value of t when Q overtakes P . [4]



One end of a light inextensible string is attached to a particle A of mass 2 kg . The other end of the string is attached to a second particle B of mass 5 kg . Particle B is in contact with a smooth horizontal surface. The string passes over a small smooth pulley P which is fixed at one edge of the surface. Particle A hangs vertically below P .

A second light inextensible string is attached to B . The other end of this second string is attached to a third particle C of mass 3 kg . This string passes over a small smooth pulley Q which is fixed at the other edge of the surface. Particle C hangs vertically below Q (see diagram).

The system is released from rest with the strings taut and B a distance 5 m from P . In the subsequent motion A does not reach P .

- (a) Determine, in terms of g , the tension in the string connecting B and C . [4]

When the particles have been in motion for 3 seconds the string connecting A and B breaks. Particle B reaches Q after a further 2 seconds.

- (b) Determine the distance PQ . [6]
 (c) State what it would mean about the tension in the string connecting B and C if pulley Q could **not** be assumed to be smooth. [1]

It is given that in fact the horizontal surface is **not** smooth. When the system, as shown in the diagram, is released, it remains at rest. (You may assume that pulley Q is smooth.)

- (d) Complete the diagram in the Printed Answer Booklet showing all the forces acting on B . [1]
 (e) State the magnitude of the frictional force acting on B . Give a brief justification for your answer. [1]

END OF QUESTION PAPER

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